# Defaults, Present Bias and Retirement Saving: A Synthetic Test Paper*

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**Abstract**

We study how default contribution rates interact with present bias in a field experiment with 1,200 employees. Automatic enrollment reduces the share of employees who save nothing by 23% relative to the control group. A present-bias parameter of $\beta = 0.82$ explains the pattern. We discuss implications for the design of retirement plans.

JEL codes: D15, D91, G51

Keywords: defaults, present bias, retirement saving, field experiment

## 1 Introduction

Many employees save too little for retirement. Present bias, the tendency to overweight immediate costs and rewards relative to later ones, is a leading explanation (Laibson, 1998; O'Donoghue and Rabin, 1999). Defaults are a powerful tool against inertia: automatic enrollment sharply raised 401(k) participation in the firm studied by Madrian and Shea (2001).

Earlier studies have also examined present bias in gym attendance (Madrian and Shea, 2001). DellaVigna and Malmendier (2006) show that gym members who choose flat-rate contracts overestimate their future attendance. Loss aversion, with a coefficient of about 2.25, was first estimated by Kahneman and Tversky (1979).

We run an RCT with a large employer. Our randomized controlled trial (RCT) varies the default contribution rate. In our previous work (Lu, 2024), we found similar patterns in a pilot. This paper contributes to a growing literature on choice architecture (Thaler and Benartzi, 2004; Smithson and Okafor, 2019; Brown, 2015).

The rest of the paper is organized as follows. Section 2 describes the design, Section 3 reports the results, and Table ?? reports robustness checks. TODO: add the heterogeneity analysis.

## 2 Experimental design

We partnered with a large employer and randomized 1,150 newly hired employees into a control group and a treatment group with automatic enrollment at a 3% contribution rate. The experiment was preregistered. Participants also completed an online survey programmed in oTree, in which they made incentivized intertemporal choices.

In the survey, participants were told that their choices would be matched with a real coworker, although all partners were simulated by the computer. We ran 12 survey sessions between March and May 2025.

Let $\beta$ denote the present-bias parameter and $\delta$ the long-run discount factor. The utility of a consumption stream is

$$U_t = u(c_t) + \beta \sum_{k=1}^{T} \delta^k \, u(c_{t+k}). \tag{1}$$

Equation (1) reduces to exponential discounting when $\beta = 1$.

The mean rating of plan clarity was 3.47 on a five-point scale ($N = 25$ pilot participants).

## 3 Results

Table 1 reports the main estimates. Automatic enrollment lowers the share of employees with zero contributions from 9.9% in the control group to 8.1% in the treatment group. The participation rate rose from 20% to 25%, an increase of 25 percentage points.

The treatment effect on the contribution rate is 0.120 percentage points (standard error 0.080), which is statistically significant at the 1% level. This correlation shows that defaults cause higher saving among present-biased employ-

*We thank seminar participants for comments. The experiment was approved by the Institutional Review Board of Example University (protocol 2024-117).
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ees.[1]

Table 1: Effect of automatic enrollment

| | (1) Zero contributions | (2) Contribution rate |
|---|---|---|
| Treatment | −0.018** | 0.120*** |
| | (0.008) | (0.080) |
| Present bias index | 0.250*** | −0.031 |
| | (0.050) | (0.042) |
| Control mean | 0.099 | 3.12 |
| Observations | 1,150 | 1,150 |

Notes: Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 1 shows the distribution of contribution rates.

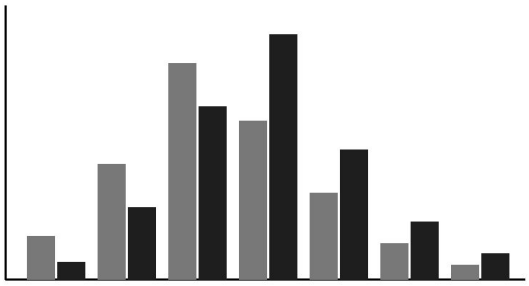


Figure 1: Distribution of contribution rates by treatment arm.

1Payments were made through the employer's payroll system within two weeks.

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# A Additional details

We ran 10 survey sessions in total. Participants earned on average $12.50, including a $5 show-up fee.